RAFFLES INSTITUTION
H2 Computing (9569)

2026 Year 6

Chapter S4B: Web App HTML

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1. A Web Page

Web pages are written using the **HyperText Markup Language (HTML)**. HTML is used to describe the structure of web pages, not to write step-by-step programs.

To view the HTML source code of a web page in Chrome, press Ctrl-U. (Alternatively, right-click and select “View page source”.) Try this now for the web page on `www.example.com`.

Examine the HTML source code on the web browser. Do you see that the contents of the web page are surrounded by text enclosed in angle brackets (i.e., < and >)?

```html
<title>Example Domain</title>
```

The text surrounded by angle brackets are special processing instructions for the web browser called **tags**.

You may also notice that the first part of the HTML document includes some code that appears to be written in another computer language. This is a **style sheet** written in the **Cascading Style Sheets (CSS)** language that controls the web page's appearance. CSS will be covered later.

2. Anatomy of a HTML Document

You can identify tags in a HTML document as portions of text that are enclosed in angle brackets. Tags can be further classified into **start tags** and **end tags**. Start tags may also have one or more **attributes**:

Start Tag Contents End Tag

```html
<a href="/example">More information...</a>
```

Attribute

Notice that end tags always start with a forward slash (/).

Certain characters have special meanings in HTML and would cause a syntax error if they are used without escaping them. In HTML, escape codes are called **character references** that start with an ampersand ‘&’ and end with a semicolon ‘;’. Some common character references are:

Character	Ampersand	Less-Than Sign	Greater-Than Sign	Quotation Mark
	&	<	>	"
Character Reference	&	<	>	"

Quiz 1

Look at the following HTML document.

```
<!doctype html>
<html>
<head><title>Welcome Page</title></head>
<body>
    <p>Welcome to our R&amp;D department.</p>
    <img src="welcome.png" alt="Welcome">
    <p>I was sent to A&E in an ambulance.</p>
</body>
</html>
```

1. Identify the start tags and end tags. Which start tags do not have matching end tags?

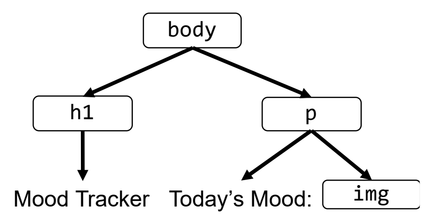
   ____

2. Identify the syntax error and correct it so that it works as intended on all browsers.

   ____

HTML tags organise content into elements. Elements can contain text and other elements.

```
<body>
    <h1>Mood Tracker</h1>
    <p>
        Today's Mood:
        <img src="happy.jpg" alt="happy">
    </p>
</body>
```



Some tags (e.g. <p>) have start and end tags. Others (e.g. <img>, <input>) do not.
Comments in HTML must start with ‘`<!--`’ and end with ‘`-->`’.

Quiz 2

For each code snippet below, place a tick in the corresponding box if the snippet is a HTML comment with no syntax errors.

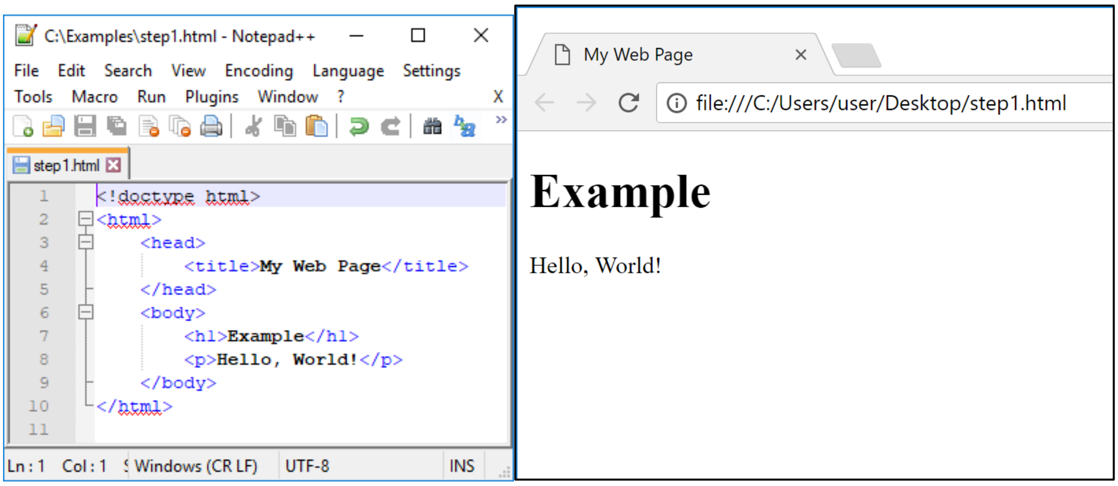
Code Snippet	Valid?
`<!-- The markup language of the web -- HTML. -->`	
`<!-- The markup language of the web - HTML. -->`	
`<!- The markup language of the web - HTML. ->`	

3. Your Own Web Page

Now is time to write your own web page. Like Python programs, HTML documents are just plain text files except that they use a `.html` extension instead of a `.py` extension. While we can write HTML using any text editor, we will use Notepad++ so that our code will be displayed with syntax highlighting. Open Notepad++, create a new file and save it in your user directory as `example.html`. Then enter the following HTML document:

```
<!doctype html>
<html>
    <head>
        <title>My Web Page</title>
    </head>
    <body>
        <h1>Example</h1>
        <p>Hello, World!</p>
    </body>
</html>
```

Save the document again and double-click `example.html` in Windows Explorer to open it in a web browser. You should see something similar to the following:



Congratulations! You have just created a web page.

Just like how Python has many functions, HTML has a vocabulary of tags such as <head>, <title>, <body>, <h1> and <p> that you should become familiar with.

We will introduce these HTML tags by organising them into five groups: *required tags*, *structural tags*, *tags for describing text and media*, *tags for describing tables* and *tags for describing forms*.

4. Required Tags

All HTML documents have a number of required elements. First, every HTML document must start with a **doctype** that looks like this:

```
<!doctype html>
```

`<!doctype html>` tells the browser the document is written in HTML.

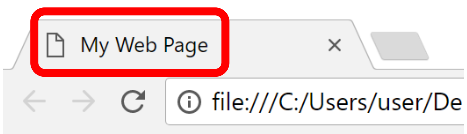
```
<!doctype html>
<html>
     ...document goes here...
</html>
```

This describes a single html element that serves at the **root element** of the document's structure. Inside the html element, there must be a head element for metadata (data that describes other data), followed immediately by a body element for the main content:

```
<!doctype html>
<html>
    <head>
        …metadata goes here…
    </head>
    <body>
        …main content goes here…
    </body>
</html>
```

Inside the head element, there must be a title element to set the title of the web page that appears in the browser window or tab:

```
<!doctype html>
<html>
    <head>
        <title>My Web Page</title>
    </head>
    <body>
        ...main content goes here...
    </body>
</html>
```

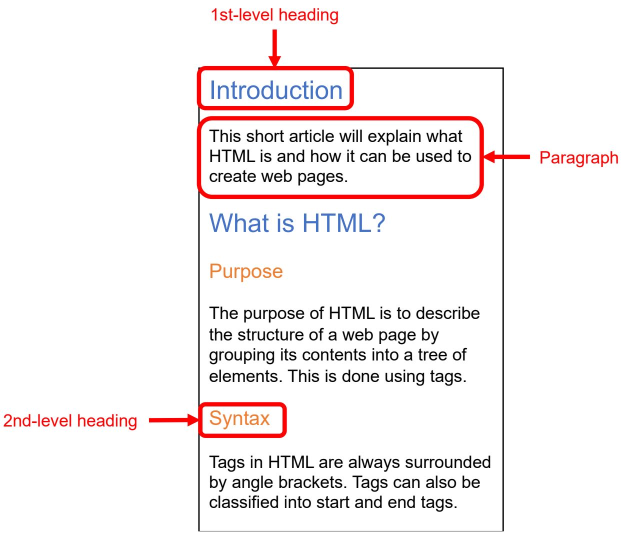


These required tags serve to ensure that all HTML documents are contained inside a single html element and that metadata is placed in a head element, separate from the main content in the body element. While `<title>` is the only metadata tag that is required inside `<head>`, we will learn about other examples of metadata that can also be included.

5. Structural Tags

Now, let us focus on the main content that goes into the `<body>` tag.

Long pieces of content are typically divided into logical parts such as headings and paragraphs.



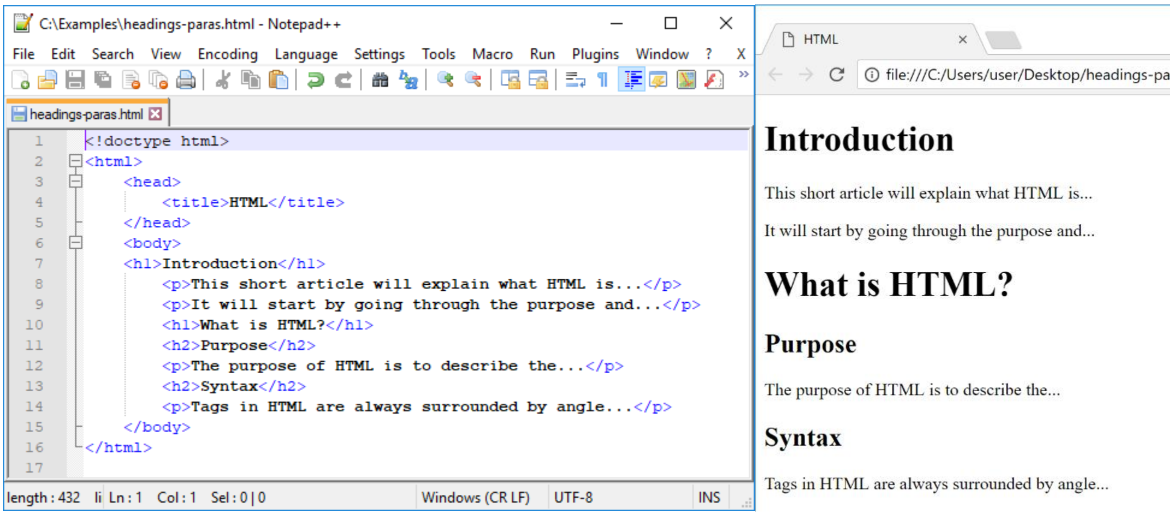
HTML provides us with 6 tags, namely `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>` and `<h6>`, to describe 1st-level headings, 2nd-level headings, 3rd-level headings and so on:

```
<body>
    <h1>Introduction</h1>
    <h1>What is HTML?</h1>
    <h2>Purpose</h2>
    <h2>Syntax</h2>
</body>
```

Content can also be organised into paragraphs using the `<p>` tag:

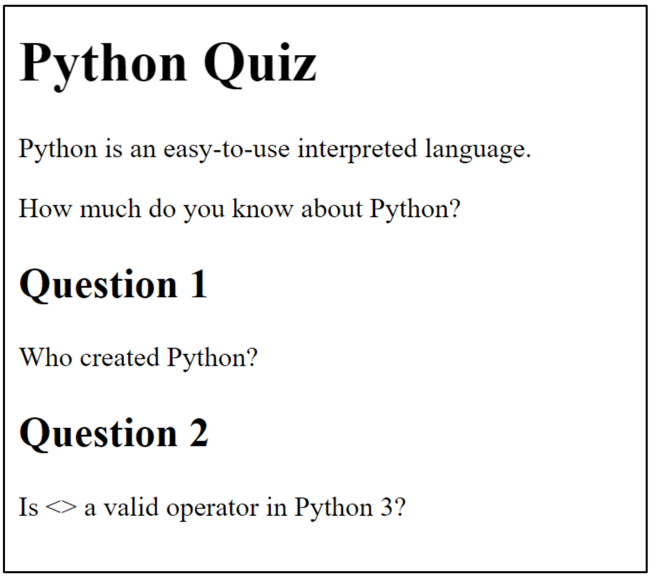
```
<body>
    <h1>Introduction</h1>
    <p>This short article will explain what HTML is...</p>
    <p>It will start by going through the purpose and...</p>
    <h1>What is HTML?</h1>
    <h2>Purpose</h2>
    <p>The purpose of HTML is to describe the...</p>
    <h2>Syntax</h2>
    <p>Tags in HTML are always surrounded by angle...</p>
</body>
```

Use a text editor to type out the above HTML, add any required tags (e.g., give the document a title of "HTML") and save the result as `headings-paras.html`. Open it in a web browser and verify that the result is similar to the screenshot on the right:



Exercise 1

Use only the `<html>`, `<head>`, `<title>`, `<body>`, `<h1>`, `<h2>` and `<p>` tags to create the following web page and save it as `quiz.html`:



In addition to headings and paragraphs, content can be organised as lists:

- Unordered list (bulleted list) using `<ul>`
- Ordered list (numbered list) using `<ol>`

Each item in a list is represented using the `<li>` (list item) tag.

Unordered list displays a bulleted list:

```
<body>
    <h1>Fruits List</h1>
    <ul>
        <li>Apples</li>
        <li>Banana</li>
        <li>Cherry</li>
    </ul>
</body>
```

Ordered list displays a numbered list:

```
<body>
    <h1>The seven habits</h1>
    <ol>
        <li>Be proactive</li>
        <li>Begin with the end in mind</li>
        <li>Put first things first</li>
        <li>Think win-win</li>
        <li>Seek first to understand, then to be understood</li>
        <li>Synergise</li>
        <li>Sharpen the saw</li>
    </ol>
</body>
```

Exercise 2

Create a HTML document named `routine.html` that displays the following content.

The page title should be **"Lists"**.

Use only the following tags:
`<html>, <head>, <title>, <body>, <h1>, <h2>, <ul>, <ol>, <li>`

My Daily Routine

Morning

- Wake up
- Brush teeth
- Eat breakfast

After School

1. Do homework
2. Revise notes
    - Math
    - Computing
3. Relax

Weekend Activities

- Sports
- Coding practice
- Family time

6. Text and Media Tags

Previously, we mentioned that CSS should be used to control the appearance of web pages. However, for basic text formatting, by convention we can also use the `<b>`, `<i>` and `<u>` tags to format text as bold, italic and underline respectively:

```
Text can be <b>bold</b>, <i>italic</i>, <u>underline</u> or
<b><i><u>all three</u></i></b>.
```

Line breaks and separators can make the page easier to read and more organised. The `<br>` tag inserts a single line break, and the `<hr>` tag creates a thematic break. Both have no end tags.

```
Line 1<br>
Line 2
<hr>
Next Section
```

Besides basic formatting, a web page may also contain images and links to other web pages. For example, the following HTML snippet describes a link to `www.example.com` using an `<a>` tag with a `href` attribute:

```
<a href="http://www.example.com">Example</a>
```

The `<a>` tag stands for "anchor" as it attaches some portion of the web page to a fixed URL, just like how a ship's anchor attaches the ship to a fixed location. The `href` attribute stands for "**hypertext reference**" and is set to the URL that `<a>` tag's content will be linked to.

Note that when linking to an external site, the `href` attribute must use an **absolute URL**. This means that the attribute value must include the scheme component of the URL (i.e., `http://www.example.com`) and cannot be written as `www.example.com`. (This is because `www.example.com` is interpreted as linking to a file named `www.example.com` and not the external site.)

If a URL does not include the scheme component, it is considered to be a **relative URL**. This means that the web browser will treat the URL as a path name starting from the web page's initial directory. For instance, the following HTML snippet describes a link to the `question1.html` web page in the `quiz` sub-directory:

```
<a href="quiz/question1.html">View Question 1</a>
```

Be aware that while Windows uses the backslash (`\`) to separate directory names in a path, URLs always use the forward slash (`/`). The double dot `".."` refers to the parent directory.

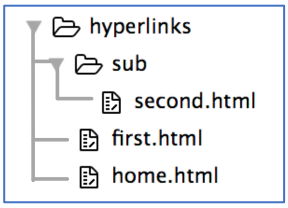
The following table shows what the resulting URL will be after following a relative URL from a given current URL of the page:

Current URL	Relative URL	Resulting URL
`http://www.sg/en`	`example`	`http://www.sg/example`
`http://www.sg/en/`	`example`	`http://www.sg/en/example`
`http://www.sg/en/sg`	`example/`	`http://www.sg/en/example/`
`http://www.sg/en/sg/`	`example/`	`http://www.sg/en/sg/example/`
`http://www.sg/en/sg`	`../example`	`http://www.sg/example`
`http://www.sg/en/sg/`	`../example`	`http://www.sg/en/example`
`http://www.sg/en/sg/`	`/example`	`http://www.sg/example`

URL resolution rules:

- **No Trailing Slash:** Drops the last segment (e.g., /en becomes /) before appending.
- **Trailing Slash:** Keeps the full path (e.g., /en/) and appends the new segment.
- **Double Dot (../):** Moves **up one level** in the directory tree before appending.
- **Leading Slash (/):** Jumps directly to the **domain root**, ignoring the current path.

Exercise 3



Create three HTML documents using the following directory structure and contents. Open `home.html` and verify that the two links to `first.html` and `second.html` work as expected.

home.html

```
<!doctype html>
<html>
    <head><title>Home Page</title></head>
    <body>
        <p><a href="first.html">First link</a></p>
        <p><a href="sub/second.html">Second link</a></p>
    </body>
</html>
```

first.html

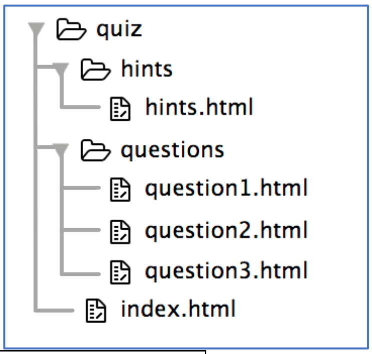
```
<!doctype html>
<html>
    <head><title>First Page</title></head>
    <body>
        <p>You are on the first page.</p>
    </body>
</html>
```

second.html

```
<!doctype html>
<html>
    <head><title>Second Page</title></head>
    <body>
        <p>You are on the second page.</p>
    </body>
</html>
```

Quiz 3

1. Given the following directory structure:



Which HTML snippet would correctly create a link from `question1.html` to `index.html`?

A	Home
B	Home
C	Home
D	Home

2. Which HTML snippet would correctly create a link from `question1.html` to `hints.html`?

A	Hints
B	Hints
C	Hints
D	Hints

3. Which of the following links to the MOE home page using the text "MOE"?

A	www.moe.gov.sg
B	http://www.moe.gov.sg
C	MOE
D	MOE

Besides links, web pages can also include images using an `<img>` tag with the `src` and `alt` attributes:

```
<img src="images/example.png" alt="Example of an image">
```

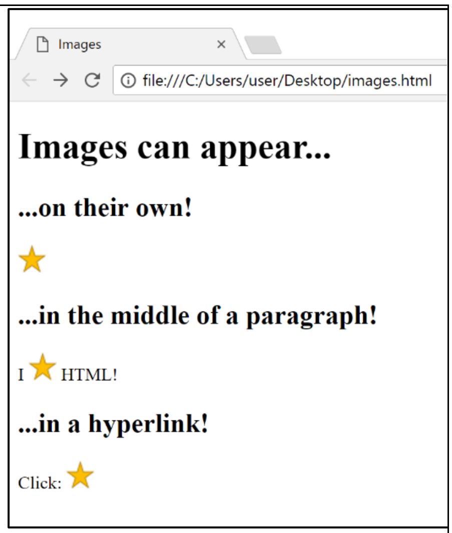
The `<img>` tag is short for "image" and its `src` attribute (short for "source") must be set to the URL of an image file. `<img>` uses **src** for file location and **alt** for alternative text.

As with the `href` attribute of the `<a>` tag, the `src` attribute of an `<img>` tag can be set to either an absolute or relative URL depending on whether the image is stored on an external site.

Exercise 4

Create a small image of a star in PNG format (using graphics software such as Paint) and save it as `star.png`. Create the following HTML document and save it as `images.html` in the same directory as `star.png`.

```
<!doctype html>
<html>
    <head><title>Images</title></head>
  <body>
    <h1>Images can appear...</h1>
    <h2>...on their own!</h2>
    <img src="star.png" alt="STAR">
    <h2>...in the middle of a paragraph!</h2>
    <p>I <img src="star.png" alt="STAR"> HTML!</p>
    <h2>...in a hyperlink!</h2>
     <p>
        Click:
        <a href="http://www.example.com">
          <img src="star.png" alt="www.example.com">
        </a>
    </p>
  </body>
</html>
```

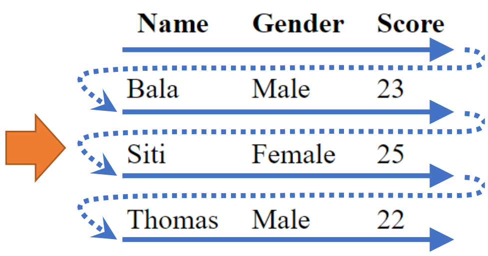


Open `images.html` and verify that the page appears as that on the right. Also check that the last star works as a hyperlink to `www.example.com`. You can also delete `star.png` and reopen `images.html` to see what happens when the required image is missing.

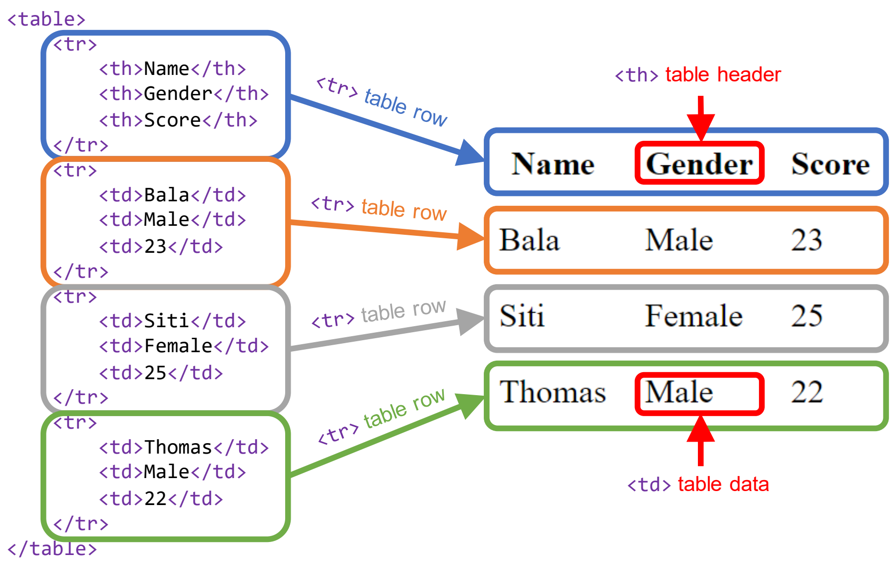
7. Table Tags

Let us describe a table using the `<table>`, `<tr>`, `<th>` and `<td>` tags:

```
<table>
    <tr>
        <th>Name</th>
        <th>Gender</th>
        <th>Score</th>
    </tr>
    <tr>
        <td>Bala</td>
        <td>Male</td>
        <td>23</td>
    </tr>
    <tr>
        <td>Siti</td>
        <td>Female</td>
        <td>25</td>
    </tr>
    <tr>
        <td>Thomas</td>
        <td>Male</td>
        <td>22</td>
    </tr>
</table>
```



`<table>` indicates the start of one table, while `</table>` indicates the end of the table. Within those tags, each row (starting from the top) is represented by a `<tr>` (table row) tag. In turn, each row contains a sequence of `<th>` (table header) or `<td>` (table data) tags, with each tag describing a cell in that table row from left to right.



Note that while both `<th>` and `<td>` are used to describe individual table cells, `<th>` should only be used for **header cells** that contain row or column titles.

Exercise 5

Create the following and save it as `time-mgt.html`:

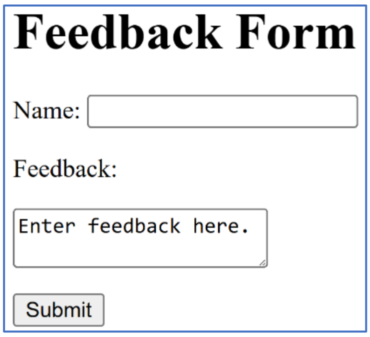
The Time Management Matrix

	Urgent	**Not Urgent**
Important	**ONE** • Crisis • Pressing problems • Deadline driven projects	**TWO** • Relationship building • Personal growth • Recreation
Not Important	**THREE** • Interruptions • Phone calls	**FOUR** • Trivia • Time wasters

Focus on **Quadrant Two**: <u>ignore</u> the ***trivial***, <u>manage</u> the ***urgent***, and <u>lead</u> your ***life***.

8. Form Tags

An important advantage of web pages over conventional documents is that web pages can collect inputs from the reader using forms such as the following example:



Such forms are created in HTML using the `<form>`, `<input>` and `<textarea>` tags:

```
<form method="POST" action="http://www.example.com">
    <h1>Feedback Form</h1>
    <p>Name: <input name="username" type="text" value=""></p>
    <p>Feedback:</p>
    <textarea name="feedback">Enter feedback here.</textarea>
    <p><input type="submit"></p>
</form>
```

Each form is contained in a separate `<form>` tag with two important attributes: **action** and **method**.

The action attribute specifies the URL where the submitted data is sent. In a web application, this is typically handled by a server-side program (e.g. a Python Flask route) which processes the data and returns a response.

The method attribute specifies how the form data is sent to the server:

- GET method:
    - Data is appended to the URL (visible in address bar)
    - Used for simple or non-sensitive data
- POST method:
    - Data is sent in the request body (not visible in the URL)
    - Commonly used for submitting user input

If the method is not specified, the default is GET.

Inside the `<form>` tag, `<input>` and `<textarea>` tags are used to collect user input. Each input control has a name attribute, which allows the server to identify the data. In Flask, values can be accessed using the name as a key (e.g. request.form["username"]).

An `<input>` tag can represent different controls depending on its type:

- "text" → single-line text box
- "submit" → submit button
- "file" → file upload

The value attribute sets the default text (for text input) or label (for submit button). The `<input>` tag does not have an end tag.

The `<textarea>` tag represents a multi-line text box and requires an end tag. The text inside it is the default content.

To make the text box more accessible and usable, the `<label>` tag can be used:

```
<label>
  Name:
  <input name="username" type="text" placeholder="Enter your name" required>
</label>
```

This code uses implicit labelling by nesting the `<input>` inside the `<label>`, automatically linking them for better usability (clicking the text focuses the box). The **name** attribute is essential for server-side data retrieval (e.g., in Flask), while **placeholder** provides a user hint and **required** acts as basic client-side validation to ensure the field isn't empty upon submission.

Exercise 6

1. Use the `<html>`, `<head>`, `<title>`, `<body>`, `<form>`, `<input>`, `<textarea>`, `<h1>` and `<p>` tags to create the following form and save it in `search.html`. (You may choose any URL for the action attribute of the form.)

2. Use the `<html>`, `<head>`, `<title>`, `<body>`, `<form>`, `<input>`, `<textarea>`, `<h1>`, `<p>` and the table tags to create the following form and save it in `game.html`. (You may choose any URL for the action attribute of the form.)